

FinWallet Happy Path - Registration to Bank-to-Wallet Funding

This document shows the actual FinWallet v1 happy path from new-customer registration through moving money from the customer's external bank account into the digital wallet. All normal customer calls go through YARP Gateway.

0. Base URL and sample variables

```
{{gateway}} = http://localhost:8080
{{token}} = login response accessToken
{{customerId}} = register response customerId
{{walletId}} = wallet create response walletId
{{bankAccountId}} = FinWallet internal BankAccount id
{{externalAccountId}} = FakeBank provider account id
{{internalKey}} = local development Gateway InternalServiceKey
```

A normal client does not call FinWallet.Api or fake-provider ports directly. The single client entry point is {{gateway}}.

1. CLIENT - Register

```
POST {{gateway}}/api/v1/auth/register
Content-Type: application/json
X-Correlation-Id: hp-register-001
```

```
{
  "countryCode": "TR",
  "phoneNumber": "+905321111111",
  "email": "happy.path@example.test",
  "password": "Example-Password-123!"
}
```

Successful response:

```
{
  "isSuccess": true,
  "code": "REGISTRATION_ACCEPTED",
  "message": "Registration accepted and verification is pending.",
  "data": {
    "customerId": "11111111-1111-1111-1111-111111111111",
    "otpExpiresAt": "2026-08-10T20:03:00+00:00"
  },
  "errors": []
}
```

FinWallet durably writes Customer and Credential state to MSSQL. The customer is not Active yet.

2. INTERNAL - OTP SMS delivery

After the registration DB transaction is complete, the handler calls FakeCommunication through Gateway.

```
POST {{gateway}}/providers/communication/api/v1/communication/sms
X-Internal-Service-Key: {{internalKey}}
Content-Type: application/json
X-Correlation-Id: hp-register-001
```

Example internal request:

```
{
  "recipient": "+905321111111",
  "messageType": "RegistrationOtp",
  "body": "FinWallet verification code: 123456",
  "correlationId": "hp-register-001"
}
```

FakeCommunication success response:

```
{
  "isSuccess": true,
  "code": "MESSAGE_ACCEPTED",
  "message": "Message accepted by fake provider.",
  "data": {
    "messageId": "22222222-2222-2222-2222-222222222222",
    "status": "Accepted",
    "acceptedAt": "2026-08-10T20:00:01+00:00"
  },
  "errors": []
}
```

This is not a normal user-facing call. The OTP is never returned in the public register response. In local tests, the sample OTP is read from simulator instrumentation.

3. CLIENT - Verify OTP

```
POST {{gateway}}/api/v1/auth/registration/verify
Content-Type: application/json
X-Correlation-Id: hp-verify-001
```

```
{
  "customerId": "11111111-1111-1111-1111-111111111111",
  "code": "123456"
}
```

Successful response:

```
{
  "isSuccess": true,
  "code": "REGISTRATION_VERIFIED",
  "message": "Registration verification completed.",
  "data": null,
  "errors": []
}
```

The Customer is now Active.

4. CLIENT - Login

```
POST {{gateway}}/api/v1/auth/login
Content-Type: application/json
X-Correlation-Id: hp-login-001
```

```
{
  "phoneNumber": "+905321111111",
  "password": "Example-Password-123!",
  "deviceId": "happy-path-device-01"
}
```

Successful response:

```
{
  "isSuccess": true,
  "code": "AUTHENTICATED",
  "message": "Authentication completed successfully.",
  "data": {
    "customerId": "11111111-1111-1111-1111-111111111111",
    "sessionId": "33333333-3333-3333-3333-333333333333",
    "accessToken": "<JWT>",
    "accessTokenExpiresAt": "2026-08-10T20:10:00+00:00",
    "refreshToken": "<OPAQUE_REFRESH_TOKEN>",
    "refreshTokenExpiresAt": "2026-08-24T20:00:00+00:00"
  },
  "errors": []
}
```

Use Authorization: Bearer {{token}} for subsequent public financial endpoints.

5. CLIENT - Create TRY Wallet

```
POST {{gateway}}/api/v1/wallets
Authorization: Bearer {{token}}
Content-Type: application/json
```

X-Correlation-Id: hp-wallet-001

```
{
  "currency": "TRY"
}
```

First successful response:

```
{
  "isSuccess": true,
  "code": "WALLET_CREATED",
  "message": "Wallet created successfully.",
  "data": {
    "walletId": "44444444-4444-4444-4444-444444444444",
    "currency": "TRY",
    "availableBalance": 0.0,
    "blockedBalance": 0.0,
    "status": "Active",
    "createdAt": "2026-08-10T20:01:00+00:00"
  },
  "errors": []
}
```

A new wallet never mints money; it starts at zero.

6. CLIENT - Open FinWallet BankAccount

POST {{gateway}}/api/v1/bank-accounts
Authorization: Bearer {{token}}
Content-Type: application/json
X-Correlation-Id: hp-bank-account-001

```
{
  "walletId": "44444444-4444-4444-4444-444444444444"
}
```

FinWallet first persists the durable internal Opening BankAccount in MSSQL. Only after the SQL transaction ends does it call FakeBank.

7. INTERNAL - Open FakeBank external account

POST {{gateway}}/providers/bank/api/v1/bank/accounts
X-Internal-Service-Key: {{internalKey}}
Content-Type: application/json
X-Correlation-Id: hp-bank-account-001

Example provider request:

```
{
  "externalCustomerReference": "11111111-1111-1111-1111-111111111111",
  "currency": "TRY",
  "requestKey": "bank-account-open:55555555555555555555555555555555"
}
```

Provider response:

```
{
  "isSuccess": true,
  "code": "BANK_ACCOUNT_ACCEPTED",
  "message": "External bank account request accepted.",
  "data": {
    "accountId": "66666666-6666-6666-6666-666666666666",
    "iban": "FWTRY6666666666666666666666666666",
    "currency": "TRY",
    "status": 2
  },
  "errors": []
}
```

status=2 means FakeBank Active.

FinWallet public response:

```
{
  "isSuccess": true,
```

```

"code": "BANK_ACCOUNT_READY",
"message": "Bank account state is available.",
"data": {
  "bankAccountId": "55555555-5555-5555-5555-555555555555",
  "walletId": "44444444-4444-4444-4444-444444444444",
  "currency": "TRY",
  "externalAccountId": "66666666-6666-6666-6666-666666666666",
  "externalIban": "FWTRY6666666666666666666666666666",
  "status": "Active"
},
"errors": []
}

```

8. TEST ONLY - Seed initial FakeBank money

A newly opened FakeBank account starts with 0 TRY. In a real bank the customer account would already hold funds. For the simulator happy path, seed the provider account through the internal test route.

This is not a normal customer API.

```

POST {{gateway}}/providers/bank/api/v1/bank/transactions
X-Internal-Service-Key: {{internalKey}}
Content-Type: application/json
X-Correlation-Id: hp-provider-seed-001

{
  "accountId": "66666666-6666-6666-6666-666666666666",
  "amount": 5000.00,
  "currency": "TRY",
  "transactionType": 1,
  "requestKey": "test-seed-bank-account-001"
}

```

transactionType=1 means provider-account Deposit, therefore credit.

Successful provider response:

```

{
  "isSuccess": true,
  "code": "BANK_TRANSACTION_ACCEPTED",
  "message": "External bank transaction request accepted.",
  "data": {
    "transactionId": "77777777-7777-7777-7777-777777777777",
    "status": 2,
    "accountBalance": 5000.00
  },
  "errors": []
}

```

9. CLIENT - Move 1,000 TRY from bank account to Digital Wallet

```

POST {{gateway}}/api/v1/bank-movements/deposits
Authorization: Bearer {{token}}
Idempotency-Key: hp-bank-to-wallet-0001
Content-Type: application/json
X-Correlation-Id: hp-bank-to-wallet-001

{
  "bankAccountId": "55555555-5555-5555-5555-555555555555",
  "amount": 1000.00
}

```

The public operation is called BankDeposit from the FinWallet point of view: money enters the wallet from the external bank.

10. INTERNAL - Debit 1,000 TRY from FakeBank account

FinWallet calls the external provider using the Withdrawal direction because money leaves the FakeBank account.

```

POST {{gateway}}/providers/bank/api/v1/bank/transactions
X-Internal-Service-Key: {{internalKey}}
Content-Type: application/json
X-Correlation-Id: hp-bank-to-wallet-001

```

```
{
  "accountId": "66666666-6666-6666-6666-666666666666",
  "amount": 1000.00,
  "currency": "TRY",
  "transactionType": 2,
  "requestKey": "888888888888888888888888888888888888"
}
```

Provider response:

```
{
  "isSuccess": true,
  "code": "BANK_TRANSACTION_ACCEPTED",
  "message": "External bank transaction request accepted.",
  "data": {
    "transactionId": "99999999-9999-9999-9999-999999999999",
    "status": 2,
    "accountBalance": 4000.00
  },
  "errors": []
}
```

After the provider reports Completed, FinWallet commits the following state in one MSSQL transaction:

Double-entry ledger posting:

If Debit and Credit are not equal, the transaction is not committed.

CLIENT response:

```
{
  "isSuccess": true,
  "code": "BANKDEPOSIT_COMPLETED",
  "message": "BankDeposit state is Completed.",
  "data": {
    "transactionId": "888888888-8888-8888-8888-888888888888",
    "bankAccountId": "55555555-5555-5555-5555-555555555555",
    "externalTransactionId": "99999999-9999-9999-9999-999999999999",
    "operation": "BankDeposit",
    "amount": 1000.00,
    "currency": "TRY",
    "state": "Completed",
    "processingDate": "2026-08-10",
    "settlementDate": "2026-08-10",
    "wasReplay": false
  },
  "errors": []
}
```

After the money commit, the Outbox worker sends the SMS. Communication failure never rolls back the financial transaction.

Example message:

{

```
"recipient": "+905321111111",
"messageType": "BANK_MOVEMENT_COMPLETED",
"body": "FinWallet notification: BANK_MOVEMENT_COMPLETED. Reference:
      88888888888888888888888888888888.",
"correlationId": "hp-bank-to-wallet-001"
}
```

When the provider accepts it, the Outbox row becomes Processed.

13. CLIENT - Verify Wallet balance

```
GET {{gateway}}/api/v1/wallets
Authorization: Bearer {{token}}
X-Correlation-Id: hp-wallet-check-001
```

Expected Wallet data:

```
{
  "walletId": "44444444-4444-4444-4444-444444444444",
  "currency": "TRY",
  "availableBalance": 1000.00,
  "blockedBalance": 0.00,
  "status": "Active"
}
```

14. CLIENT - Verify Transaction history

```
GET {{gateway}}/api/v1/transactions?take=20
Authorization: Bearer {{token}}
```

History contains the BankDeposit / Completed / 1000 TRY row together with the internal BankAccountid and external transaction reference. Raw ledger entries, passwords, tokens and sensitive provider payloads are not returned by the public history API.

Fraud note

There is no fraud-provider call in this BankDeposit happy path. In the current FinWallet v1 implementation, internal + external fraud runs before WalletTransfer and Purchase. The fraud flow is documented in 22-fraud-path.